Project Design Phase

Solution Architecture

Date	12 October 2026
Team ID	SWUID2026402491
Project Name	IMovies - Online Movie, Event & Entertainment Ticket Booking System
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for **IMovies** ensures a robust, user-friendly and scalable platform that connects audiences with cinemas and event organizers. It follows a layered three-tier design: a React single-page application, a stateless Spring Boot REST API secured with JWT and role-based access, and a relational database, supported by Redis for seat locks and caching. The architecture focuses on responsive design, conflict-free seat booking, secure mock payments and timely notifications to ensure a smooth booking journey.

- Seamless movie and event discovery, show selection and ticket booking
- Real-time seat availability with atomic seat locking to prevent double booking
- Scalable user authentication and role-based authorization (customer, manager, organizer, admin)
- Secure and trackable payments, cancellations and refunds through a mock gateway
- Reliable QR e-tickets, in-app and email notifications, and admin analytics

Example - Solution Architecture Diagram:

